DO NOW — RECALL

Lesson 136 · Normalisation: First and Second Normal Fo

Name: _____ Class: _____ Date: _____

1. What table resolves a many-to-many database relationship?

2. What is a composite key?

3. When does a linear search stop?

4. In Big O notation, what is binary search worst-case complexity?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A junction/associative table.

2. Model answer

A key made from two or more attributes together.

3. Model answer

On a match or after examining all items.

4. Model answer

O(log n) for sorted random-access data.

By the end, you can…

• Replace 'Maths, Art' in one cell with separate enrolment records.

• Check whether key is composite, such as StudentID+CourseID.

• Move StudentName to Student and CourseTitle to Course.

Vocabulary — know these terms

1NF — Attributes hold atomic values and rows can be uniquely identified.

2NF — In 1NF, no non-key attribute depends on only part of a composite candidate key.

partial dependency — Non-key field depends on a subset of a composite key.

anomaly — Insertion, update or deletion causes undesirable effects.

NEW LEARNING

Read, annotate and explain

Make fields atomic

Replace 'Maths, Art' in one cell with separate enrolment records.

Identify keys

Check whether key is composite, such as StudentID+CourseID.

Remove partial dependencies

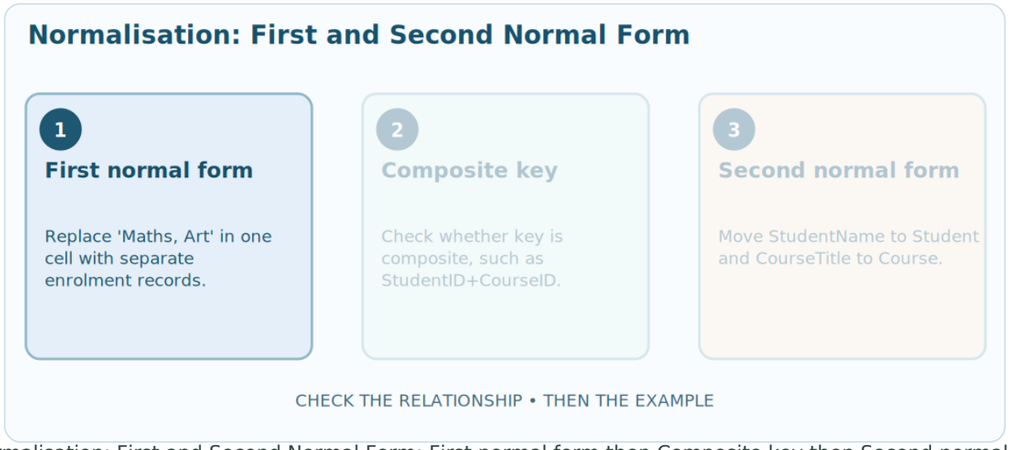
Move StudentName to Student and CourseTitle to Course.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

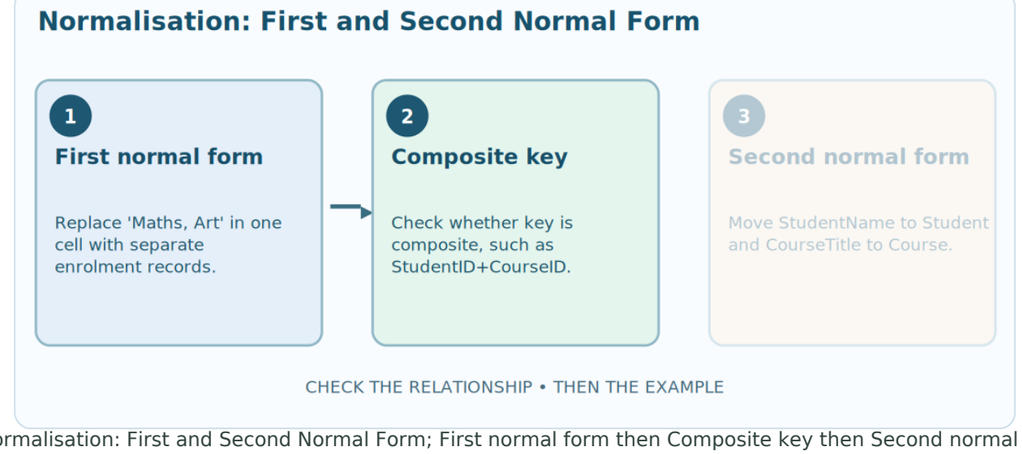
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Normalisation: First and Second Normal Form; First normal form then Composite key then Second normal form

New Learning



Normalisation: First and Second Normal Form; First normal form then Composite key then Second normal form

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

ProductName depends on ProductID

Explain why this intermediate stage is necessary.

Move name to Product

ProductName depends on ProductID alone, not whole OrderID+ProductID key. Create Product(ProductID PK,ProductName) and retain OrderLine(OrderID,ProductID,Qty).

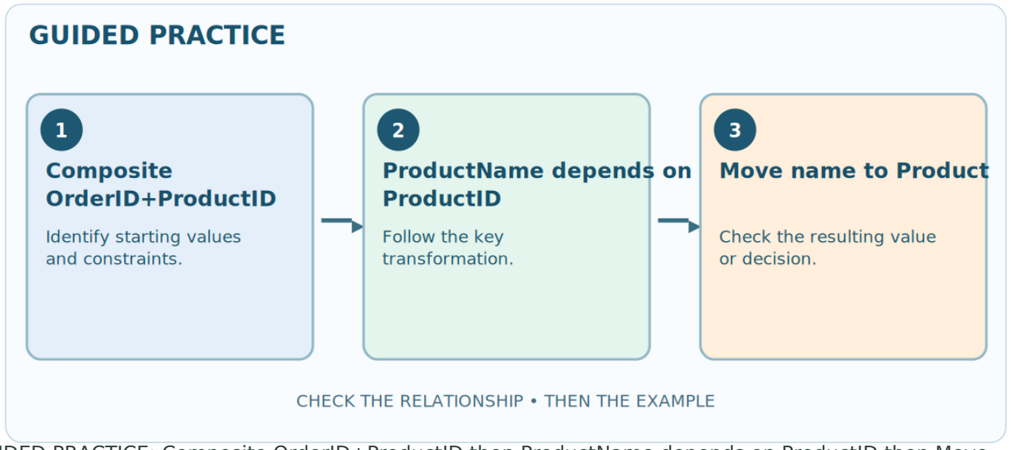
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

OrderLine(OrderID,ProductID,ProductName,Qty) uses composite PK. Identify partial dependency and fix.

Find the given data and name the required result.

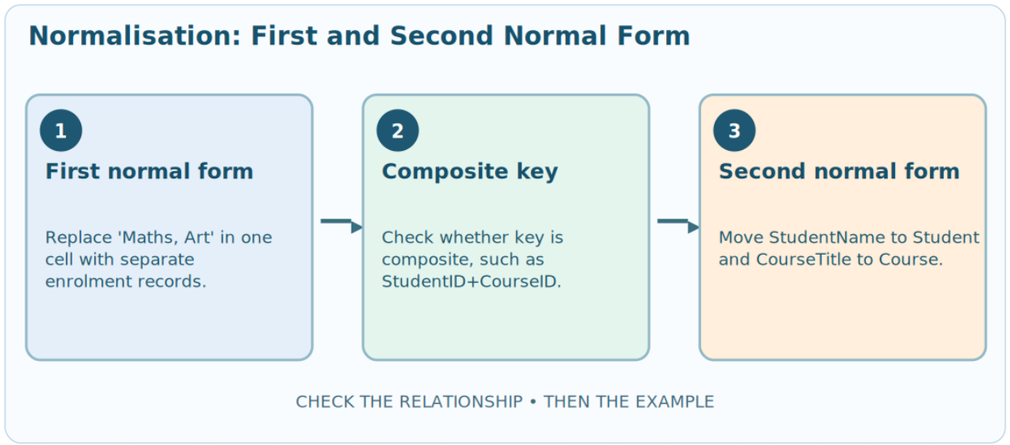


GUIDED PRACTICE; Composite OrderID+ProductID then ProductName depends on ProductID then Move name to Product

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Normalisation: First and Second Normal Form; First normal form then Composite key then Second normal form

I DO — WORKED EXAMPLE

Follow the worked method

Enrolment(StudentID,CourseID,StudentName,CourseName,Grade) has composite PK (StudentID,CourseID). Achieve 2NF.

StudentID+CourseID key

Find the given data and name the required result.

Names partly depend on key

Explain why this intermediate stage is necessary.

Split Student and Course

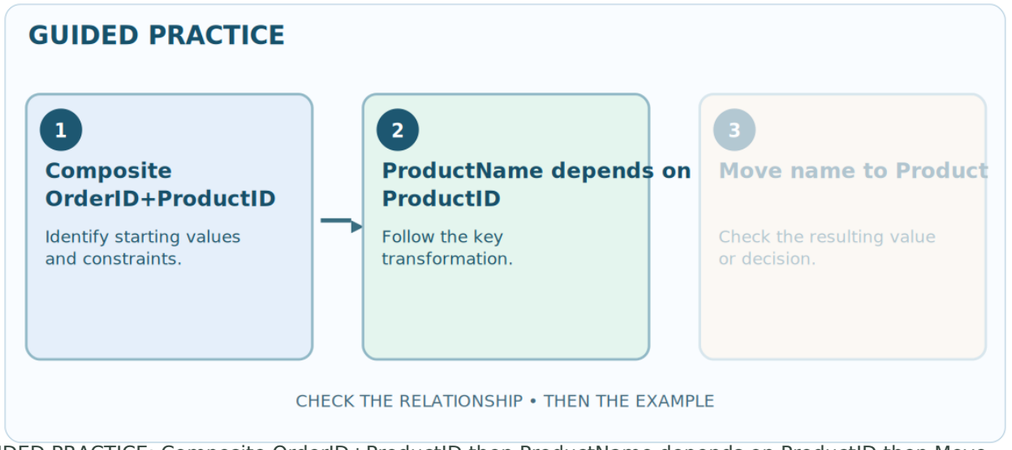
StudentName depends only on StudentID; CourseName only on CourseID. Split into Student(StudentID PK,StudentName), Course(CourseID PK,CourseName), Enrolment(StudentID FK,CourseID FK,Grade) with composite PK.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

OrderLine(OrderID,ProductID,ProductName,Qty) uses composite PK. Identify partial dependency and fix.

Find the given data and name the required result.



GUIDED PRACTICE; Composite OrderID+ProductID then ProductName depends on ProductID then Move name to Product

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A relation with a single-attribute candidate key can have a partial dependency on that key.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

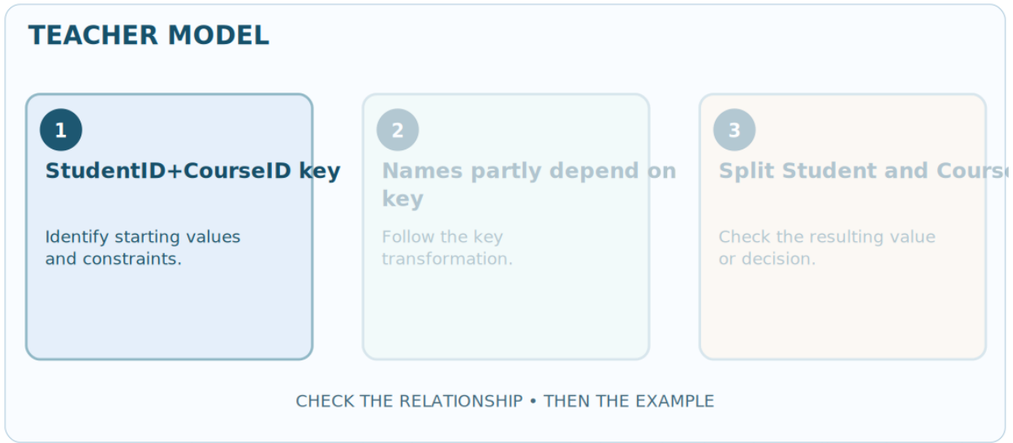
Next step: revisit the model or a relevant revision-guide section if needed.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Enrolment(StudentID,CourseID,StudentName,CourseName,Grade) has composite PK (StudentID,CourseID). Achieve 2NF.

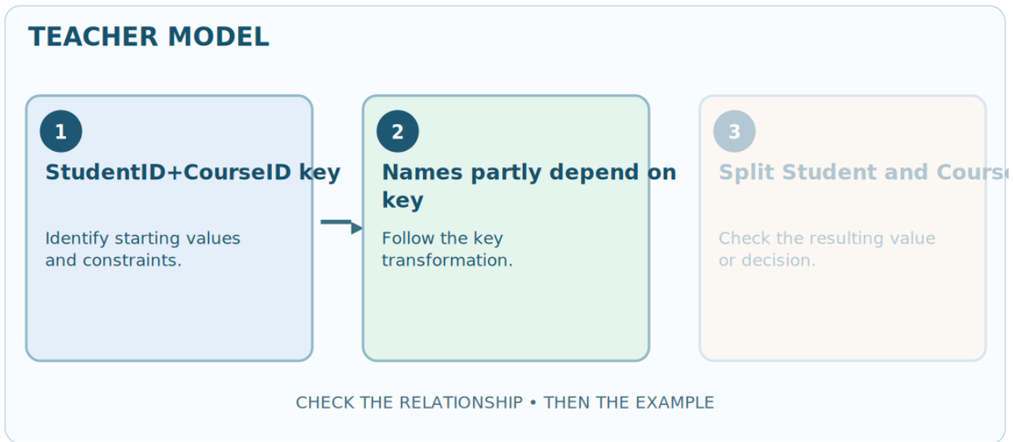
Find the given data and name the required result.



TEACHER MODEL; StudentID+CourseID key then Names partly depend on key then Split Student and Course

Enrolment(StudentID,CourseID,StudentName,CourseName,Grade) has composite PK (StudentID,CourseID). Achieve 2NF.

Find the given data and name the required result.



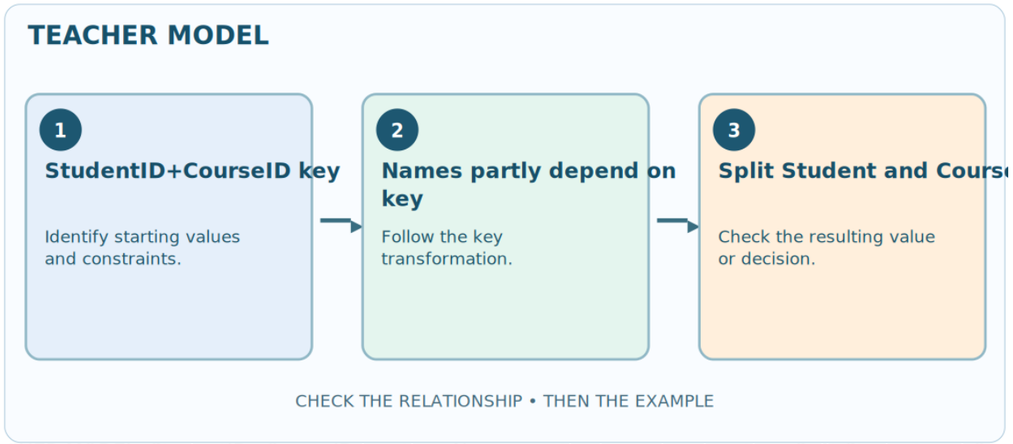
TEACHER MODEL; StudentID+CourseID key then Names partly depend on key then Split Student and Course

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Enrolment(StudentID,CourseID,StudentName,CourseName,Grade) has composite PK (StudentID,CourseID). Achieve 2NF.

Find the given data and name the required result.



TEACHER MODEL; StudentID+CourseID key then Names partly depend on key then Split Student and Course

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Convert StudentID,ClubID,StudentName,ClubName,JoinDate with composite key to 2NF. [8]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain atomicity in 1NF and correct a cell containing 'Red, Blue, Green'. [6]

Question 2

Explain partial dependency using OrderID+ProductID key and ProductDescription. [6]

WE DO — GUIDED PRACTICE

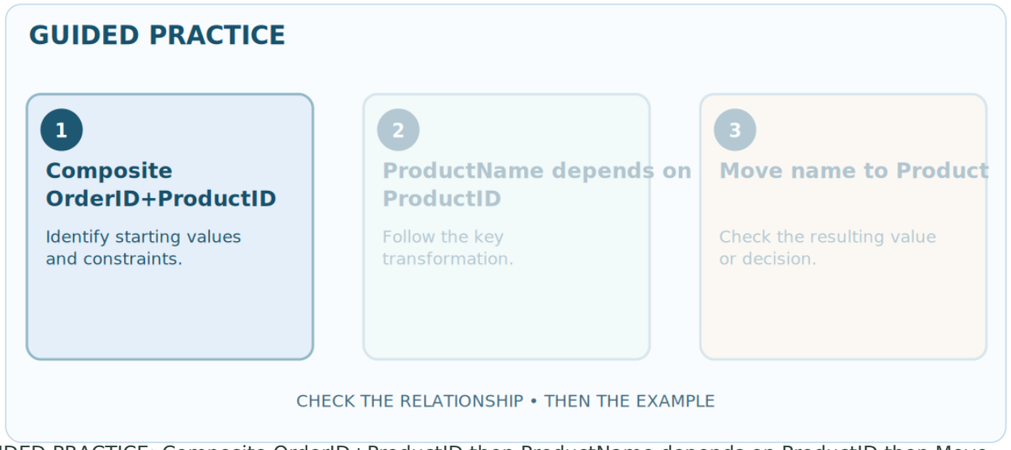
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

OrderLine(OrderID,ProductID,ProductName,Qty) uses composite PK. Identify partial dependency and fix.

Find the given data and name the required result.



GUIDED PRACTICE; Composite OrderID+ProductID then ProductName depends on ProductID then Move name to Product

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Composite OrderID+ProductID Find the given data and name the required result.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.